# Conduction vs Insulation by RealQM

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**Abstract**

This article shows what RealQM, as a new alternative to textbook quantum mechanics, offers as explanation of the difference between conductors and insulators. RealQM represents an $N$-electron system in terms of $N$ real-valued wave functions $\psi_i$, whose squares $\psi_i^2$ are unit charge densities with non-overlapping supports meeting at free interfaces. The physics is described through an energy functional, the sum of electron Coulomb potential energy without self-repulsion and kinetic energy, with no other force entering and nothing fitted; the ground state minimises it over the supports as well as over the wave functions, so the interfaces are unknowns of the problem. The cost is polynomial of small power in $N$.

Computationally the difficulty is one of scale. Atomic energies are tens of electron-volts; semiconductor activation is 50–500 meV; metallic conduction sets in near 20 meV. Against the total energy of a system large enough for the question to be posed those are $10^{-3}$, $10^{-5}$ and $10^{-6}$, so whether a material conducts is decided by a difference five or six orders of magnitude below the number a calculation returns. The same span falls on the model: a description adequate for binding, where a per cent error is invisible, can be hopeless for conduction, where a per cent of the total is a hundred times the quantity in question.

Two models are measured. A **chain of atoms**, the valence electron occupying its own kernel's region, gives a corrugation of order an electron-volt for the electron pattern to slide through, and an extensive polarisability $\alpha = 27.4N$: an insulator. A **cable of ions surrounded by electrons**, the carriers standing off cores that screen the kernels, gives ≈240 meV — the semiconductor range, where transport is thermally activated hopping. What sets that barrier is less the ion lattice than how strongly the carriers resist being modulated by it, and RealQM's free interface resists very little. Doubling the coefficient of the kinetic energy, from the $\frac{1}{2}$ of atomic standardisation to 1, tests how much turns on that, and gives ≈24 meV: an elevenfold fall, to thermal energy. The value is not calibrated to anything — it is one point, showing that a little carrier stiffness is enough to remove the pinning altogether.

The corrugation is the activation energy $E_a$ in a hopping conductivity $\sigma \propto \nu\, e^{-E_a/kT}$. Closing the prefactor with an attempt frequency $\nu = (1/2\pi)\sqrt{K/M}$, built from the curvature of that landscape and an ionic mass, gives $\nu \approx 10^{12}$–$10^{13}$ Hz, a lattice frequency, and room-temperature conductivities of $2 \times 10^{-12}$ and $6.5\ \mathrm{S\,m^{-1}}$ for the first two — glass and germanium, with nothing fitted. For the third the expression does not apply, a barrier at $kT$ being no barrier: that state is unpinned, and what it is instead is left open. Only $E_a$ is measured here; the mass and the carrier density are inputs, since RealQM carries no masses. A resistive conductivity is outside in any case, needing a relaxation time from electron–phonon scattering.

## 1 The framework, and what is asked of it

This article shows what RealQM, as a new alternative to textbook quantum mechanics, offers as explanation of the difference between conductors and insulators.

RealQM represents an $N$-electron system in terms of $N$ real-valued wave functions $\psi_i$, whose squares $\psi_i^2$ are unit charge densities with non-overlapping supports $\Omega_i$ meeting at free interfaces. The physics of the system is described through its energy functional, the sum of electron Coulomb potential energy without self-repulsion and kinetic energy, with no other force entering and nothing fitted. Explicitly,

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$. The ground state of the system is the configuration minimising this energy, the minimisation running over the supports as well as over the wave functions, so that the interfaces are unknowns of the problem and not inputs to it; excited states are the other stationary points of the same energy. The cost of minimising (1) is polynomial of small power in $N$.

Every previous test of RealQM — ionisation energies across the periodic table, molecular binding, hydrogen bonds and solvation, nuclear binding by dual Coulomb confinement — has been on *bound* matter, where an attractor is always present and the charge is where the attractor is. Conduction is the first question that asks about charge which is not where an attractor holds it, and the framework has no separate machinery for it: the same energy, the same domains, in a different arrangement of the same two kinds of charge.

**Two arrangements, both ordinary matter.** The first is a **chain of atoms**: one kernel per site with its valence electron occupying the same region, which is what a hydrogen chain is and what an alkali is to a first approximation. The second is a **cable of ions surrounded by electrons**: kernels screened by their own core electrons, with the valence charge standing off outside them, which is how a wire is built. The question is what each gives.

## 2 What textbook quantum mechanics offers

Recall that textbook quantum mechanics is based on a single antisymmetric wave function $\Psi$ of the $N$ electron coordinates $x_1,\dots,x_N$, hence on $3N$ dimensions, with the exclusion principle carried by the antisymmetry and a cost exponential in $N$; the methods actually used — Hartree, Kohn–Sham and their descendants — proceed by a drastic reduction of that dimensionality, and to that extent are no longer determined by Coulomb interaction and kinetic energy alone.

For conduction in particular there is then no single standard theory but a stack of them, each carrying quantities supplied from measurement. Metal is separated from insulator by a gap at the Fermi level [1], and in practice the gap comes from a chosen exchange–correlation functional [2]; the Kohn–Sham gap is not the excitation gap even for the exact functional [3], and the discrepancy is repaired by hybrid mixing or a $GW$ self-energy [4]. Transport is then a second model on top of the first: a Boltzmann equation closed by a relaxation time, with an effective mass and a deformation potential for the semiconductor [5]; a transfer integral and a reorganisation energy for a hopping conductor [6, 7]; a localisation length and a density of states at the Fermi level for variable-range hopping in a disordered one [8]; a Hubbard $U$ where the band picture fails outright and the material insulates by correlation instead [9, 10]. The point of the contrast is not accuracy — these are accurate, and are what one should use to predict a number for a given material. It is that they are several models with parameters, addressing the three regimes separately, whereas what follows is one energy with none, in which the regimes differ by how the charge is arranged.

Equation (1) has no $3N$-dimensional object in it and no antisymmetry: the role played by the exclusion principle is played instead by the requirement that the supports do not overlap.

## 3 What is measured

A domain carries one unit of charge by constraint, so no charge passes between sites. What can move is the *partition*: the boundaries are redrawn, and the pattern of domains advances through the lattice. When the pattern has advanced by one full spacing, every electron has taken over its neighbour's kernel and the configuration is restored — identical, since the electrons are identical and the energy depends on the partition and not on labels. Charge has been transported and nothing material has traversed the sample, which is the relation a dislocation bears to slip.

The quantity that decides whether this happens is the energy along that path. Write $E(d)$ for the energy with the partition displaced by a fraction $d$ of a spacing. It is periodic, $E(d+1) = E(d)$, and its amplitude

$$V_0 = \max_d E(d) - \min_d E(d) \tag{2}$$

is the corrugation: the barrier the partition must cross to advance one site. A field tilts this curve by $-NeFd$, so the pattern runs when the tilt exceeds the steepest slope of $E(d)$, and otherwise settles at a displacement and stops — bounded polarisation, no current.

**Ends, and how to remove them.** A finite chain has ends, and they dominate: terminal domains bind an electron-volt harder than interior ones, so a corrugation read from total energies is reading the ends. Displacing only the middle $M$ carriers repairs this. The energy difference is then $M$ times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly, and no terminal domain moves in either configuration:

$$\Delta E(M) = M\,\varepsilon_{\text{bulk}} + 2\varepsilon_{\text{junc}}, \qquad \varepsilon_{\text{bulk}} = \frac{\Delta E(M_2) - \Delta E(M_1)}{M_2 - M_1}. \tag{3}$$

The decomposition asserts linearity in $M$, which a third window tests rather than fits.

**Why a difference this small is available at all.** The barrier sought is some tens of meV against total energies of order a hundred hartree — five or six significant figures down — and the natural objection is that no calculation resolves that. The objection would be right if the total energy had to be accurate. It does not. What is required is that the *difference* be accurate, and the two configurations entering it are the same system: the same domains, the same electrons, the same kernels, the same total charge, on the same grid, differing only in where a few interior walls sit. Every large contribution to the discretisation error — the nuclear regions, the cores, the box — is identical in both and subtracts out. This is the ordinary reason binding energies are quoted in meV while the total energies they come from carry errors a thousand times larger.

That the cancellation actually occurs is not assumed but checked, in four independent ways. The solver is deterministic: repeated runs of one configuration agree to the last reported digit, so there is no stochastic floor beneath the quantity being measured. Translation by one full period is an exact relabelling and must return the same energy, which it does to nine digits — two separately relaxed states agreeing to that precision. The junction cancellation in (3) is exact algebra rather than an approximation, so the instrument introduces no error of its own. And the linearity the decomposition asserts is tested against a third window, which it could fail and does not. What the last of these does not do is prove the result correct; symmetry controls verify self-consistency, and a systematic error respecting the lattice symmetry would pass all of them. They establish that the arithmetic is sound, which is a smaller claim than the result being right, and a necessary one.

## 4 Model I: a chain of atoms

Seven model atoms in a row, one valence electron each, the electron occupying its kernel's region. Two measurements.

**The polarisability is extensive.** With the partition held fixed and a uniform field applied, the induced dipole gives $\alpha$, and repeating at three chain lengths gives its scaling: $\alpha = 27.4N$, flat to 1% per atom. That is the signature of charge fixed per site. A metal's polarisability grows faster than the number of atoms, because charge redistributes over the whole sample; here each domain carries its unit and cannot give it up, so the only responses are the density deforming inside a fixed domain and the boundaries displacing. Both are per-atom constants, and the sum is exactly proportional to $N$. The one escape is a vanishing corrugation, which would let the partition slide freely and the response grow without bound.

**The corrugation does not vanish.** It is of order an electron-volt per electron, and it stays there: repeating the measurement across lattice spacings from 3.5 to $8.0\,a_0$ gives a minimum near the equilibrium spacing and a rise on both sides, with no spacing approaching zero and no flip of the registry preference. Compressed, the cores crowd the electron; expanded, sharing a kernel costs more than owning one.

An electron-volt is forty times thermal energy at room temperature and lies above the 10–500 meV of the activated conductors this framework might otherwise describe. On this model the answer is unambiguous: an insulator, and not a marginal one.

## 5 Model II: a cable of ions surrounded by electrons

The chain places the carrier where the attractor is. Conducting matter does not: the core electrons occupy the region around each nucleus and the conduction charge stands outside them. Representing that needs two populations — cores held inside a radius, carriers outside it, with only the carriers free to slide — and it is the one arrangement that gives a carrier a standoff without violating the tiling, because the inner region is genuinely owned rather than left empty.

Sixteen sites, kernels of charge +3 on the axis, a closed core of two electrons at each, and one valence electron per site outside them: neutral, with no pseudopotential anywhere, so the exclusion of the carrier from the core region is by non-overlap rather than by an imposed condition. The core pair is treated as a single homogenised domain of charge two rather than as two unit domains meeting at a plane — a closed shell is spherically symmetric, and a core split into halves lets the halves relax unequally, which the eigenvalue control catches. Its self-repulsion is reduced to match: for a homogenised domain of charge $q$ the fraction $1/q$ of its own potential is removed, leaving $(q-1)/q$ of the naive self-energy, which is exactly the $\binom{q}{2}$ pairs the $q$ electrons form. At $q = 2$ that is one pair, and at $q = 1$ it removes the self-potential entirely, recovering the usual absence of self-interaction. The carriers, which are what the measurement concerns, are $q = 1$ domains throughout. The corrugation, measured by (3) with windows of four and eight carriers and checked against a third at twelve:

$$\varepsilon_{\text{bulk}} \approx 240 \text{ meV per carrier.}$$

That is inside the 10–500 meV band of the doped semiconductors, organic conductors and polaronic oxides — the class of materials whose transport is thermally activated hopping, and the class this framework has a claim on. The same interaction, the same energy, the same domains: only the arrangement of the charge has changed, and the barrier has fallen by a factor of four.

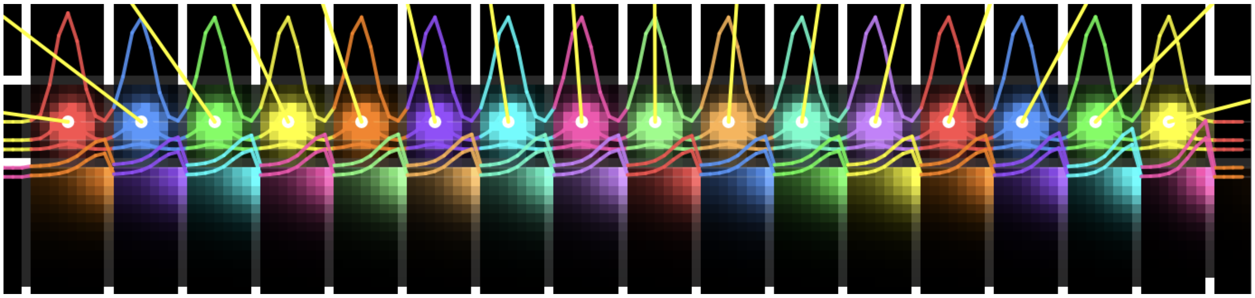

Figure 1: The cable, relaxed. Sixteen sites along the axis, each carrying a kernel of charge +3, a core domain of two electrons confined within $c_r = 2.4\, a_0$, and one valence electron outside it. Colour distinguishes domains: the bright lobes on the axis are the cores, the fainter blocks beneath them the valence charge standing off. White verticals mark the free boundaries between neighbouring carriers, and the coloured curves are density profiles along a line through the axis; the yellow segments are net forces on the kernels. Every boundary shown is an unknown of the minimisation, not an imposed partition. Displacing the middle $M$ of these carriers by half a spacing, at fixed cores and kernels, is the measurement of §2.

## 6 What sets the pinning

The barrier is not set by the strength of the ion lattice but by whether the carriers bunch into its wells. A carrier distribution that resisted being modulated would ride over the lattice; one that does not resist locks to it.

RealQM's free interface resists very little, and this is internal to the construction. A flat density is admissible on a free boundary at no cost — a constant $\psi_i$ satisfies the normalisation, has $\nabla\psi_i = 0$, and minimises the electrostatic energy against a neutralising background — so redrawing an interface between domains of equal density costs nothing. The partition therefore offers no resistance to uniform compression, and the only penalty on a *modulated* density is the gradient term in (1), carrying the coefficient $\frac{1}{2}$.

**That coefficient is a standardisation, not a constant of the theory.** It is set by calibrating the framework against atomic spectra, hydrogen first among them — on a bound electron in a single potential well. An extended arrangement is a different object, as an effective mass in a solid differs from the free-electron mass without anyone regarding it as tampering, and the coefficient governing how a carrier lattice resists modulation by an ion lattice has no obligation to equal the one an atom's spectrum sets.

So the natural thing to ask is how much the answer turns on it, and that is all the following is: a sensitivity, not a calibration. The coefficient is doubled — a round factor, chosen for being round — and the corrugation is measured again. No value was sought that would reproduce a conductivity, a barrier, or any other measured quantity, and none of the numbers below was known in advance. Doubled, with the ions, the standoff, the geometry and the mesh unchanged:

$$\varepsilon_{\text{bulk}} \approx 24 \text{ meV per carrier},$$

a fall by a factor of eleven, to thermal energy at room temperature. That is the measurement, and it is worth being careful about what it does and does not say. What it says is that the pinning no longer holds: a barrier at $kT$ does not confine, and the charge is free to move. What state the system is then in is not determined here. It is not band conduction, since non-overlap admits no extended states for a band to be made of — but that rules out a mechanism, not a phenomenon, and whether the unpinned state would behave as a metal, as a sliding charge lattice, or as something without a name in the existing vocabulary is a question this measurement does not reach. The instrument measures a barrier. When the barrier goes away, so does the instrument.

| model | corrugation per carrier | regime |
|---|---|---|
| chain of atoms: carrier on its own kernel | $\sim$1 eV | insulator |
| cable of ions: carriers standing off the cores | $\approx$240 meV | semiconductor: activated hopping |
| the same cable, carriers stiffened | $\approx$24 meV | unpinned; regime not classified |

## 7 What this gives, and what it does not

In the activated regime the corrugation is the activation energy, and the conductivity of a hopping conductor is

$$\sigma = \frac{ne^2a^2\nu}{kT}\, e^{-E_a/kT}, \qquad \nu = \frac{1}{2\pi}\sqrt{\frac{K}{M}}, \tag{4}$$

$a$ the hop distance and $n$ the carrier density. The two factors do separate work. The exponential is the fraction of the time the system has enough thermal energy to be over the barrier; the prefactor $\nu$ is how often it asks. In activated rate theory [11] a system sits near the bottom of a well, oscillates there at the small-oscillation frequency $\nu = (1/2\pi)\sqrt{K/M}$ set by the curvature $K$ of the well and the mass $M$ of what oscillates, and each oscillation is one attempt at the barrier — whence *attempt frequency*.

What oscillates here is the collective coordinate $u$: the displacement of the carrier pattern along the chain, the same variable $E(u)$ was measured against in §2. The attempt is therefore the charge pattern rocking back and forth in one corrugation well against the ion lattice, and since it is the sites that move, that rocking is a lattice vibration. Both $V_0$ and $M$ are then taken per carrier, which is consistent: displacing $m$ carriers rigidly costs $m$ times the barrier and moves $m$ times the mass, and $\nu$ is unchanged.

The measurements supply $E_a$, which decides the temperature dependence and the ratio between regimes. The prefactor needs $K$, $M$ and $n$, and it is worth being exact about where each comes from.

$K$ is the curvature of the landscape whose amplitude was measured. Taking the periodic curve to be $E(u) = \frac{1}{2}V_0\big[1 - \cos(2\pi u/a)\big]$ gives $K = 2\pi^2V_0/a^2$, which for $V_0 = 240$ meV at $a = 4.5\,a_0$ is 0.84 eV Å$^{-2}$. This is the one step that assumes a shape rather than measuring one: the windowed slide was run at the two endpoints $d = 0$ and $d = \frac{1}{2}$, which fix the amplitude and not the curvature. A scan at small $d$ would replace the assumption, and until it is run the sinusoid stands in for it.

$M$ is not a RealQM quantity at all. The framework carries no masses: the coefficient of $|\nabla\psi_i|^2$ is a geometric scale fixed by atomic standardisation, not $\hbar^2/2m$, which is precisely why §5 could raise it. So $M$ enters from outside as the mass of whatever moves — and what moves here is the site, since the charge does not pass between sites but the pattern of domains advances through the lattice. $M$ is therefore a lattice mass, and $\nu$ comes out a phonon frequency. This is a genuine seam: RealQM fixes the exponent, and one number for the prefactor is imported.

$n = 1/a^3$ takes the cable's spacing to hold in three dimensions, one carrier per site. The cable is a single line of sites, so this is a stated convention for converting a per-carrier barrier into a bulk conductivity, not something the calculation contains.

With $M = 28\,u$ and $T = 300$ K:

| | $E_a$ | $\nu$ (Hz) | $\sigma$ (S/m) | comparable to |
|---|---|---|---|---|
| chain of atoms | $\sim$1 eV | $5.5 \times 10^{12}$ | $2 \times 10^{-12}$ | glass, $10^{-11}$–$10^{-15}$ |
| cable of ions | 240 meV | $2.7 \times 10^{12}$ | 6.5 | intrinsic Ge, 2; lightly doped Si |
| cable, stiffened | 24 meV | $8.5 \times 10^{11}$ | $(8.8 \times 10^3)$ | — see below |

The attempt frequencies are $10^{12}$–$10^{13}$ Hz, which is where lattice vibrations live: the middle row is 2.7 THz, or $\hbar\nu = 11$ meV, or 90 cm$^{-1}$, an ordinary low-lying phonon. Nothing was arranged to make that so, and it is the check that the mass entering $\nu$ is the right kind of mass. The first two rows fall in the insulating and semiconducting ranges.

The third is bracketed because the formula has been carried outside the regime it describes. Equation (4) is the rate of escape from a well, and presumes a barrier large against $kT$ so that escape is rare and the prefactor counts attempts. At $E_a = 24$ meV against $kT = 25.9$ meV, $e^{-E_a/kT} = 0.4$: nothing is confined and nothing is being attempted. The number is what the hopping expression returns when extrapolated to a barrier that has ceased to exist, and it should be read as a marker for *unpinned* rather than as a conductivity. What the unpinned state actually is, and what rate law governs it, this calculation does not determine.

These are order-of-magnitude statements and cannot be more. The $\pm 30$ meV on $E_a$ is already a factor 3.2 either way in $\sigma$ at room temperature, the sinusoid is an assumption, and $M$ and $n$ are inputs. What the framework delivers is the exponent; that the resulting numbers land in the three named regimes, in the right order, with nothing fitted, is the result.

**The kink, and why the table is a lower bound.** Every barrier above is a rigid slide: all carriers displaced together, so the cost is $N$ times a per-carrier barrier and the whole sample must climb at once. Real conductors do not do this. The cheap path is a kink — one carrier advancing while the rest hold still, the domains ahead of it compressed and those behind expanded — a localised defect whose energy does not grow with the length of the chain and which, once formed, moves by shifting its centre through the lattice. This is the relation a dislocation bears to sliding a whole crystal plane, and it is the same object a charge-density wave transports charge with [14].

The carriers-on-a-substrate system of §5 is a Frenkel–Kontorova chain [12]: a harmonic coupling $\kappa$ between neighbouring carriers on the periodic potential of curvature $K$ measured here. Whether such a chain pins at all is the Aubry transition [13], controlled by the same ratio of the two constants. Two consequences follow, and both bear on the numbers above. First, the barrier changes character. A kink of width $w = \sqrt{\kappa/K}$ sites crosses a Peierls–Nabarro barrier $E_{\rm PN}$ which falls steeply — exponentially, in the continuum limit — as $w$ grows, so a wide kink slides almost freely over a substrate that pins a rigid chain hard. The conductivity is then governed by $E_f + E_{\rm PN}$, the cost of creating a kink plus the cost of moving it, in place of $E_a$; since $E_{\rm PN} \ll V_0$ the exponent falls and every $\sigma$ in the table is an underestimate, by a factor this article does not determine.

Second, and to the question of a clock: the kink is better supplied with one than the rigid slide is. The same two constants give the carrier lattice a sound speed $c_0 = a\sqrt{\kappa/M}$, which is the kink's limiting velocity, and a kink mass set by $\kappa$, $K$ and $M$ — so the rate of kink transport is fixed by $\kappa$, $K$, $E_f$, $E_{\rm PN}$ and $M$, of which only $M$ comes from outside. The width $w$ in particular is a pure ratio of two curvatures and therefore wholly a RealQM quantity. Nothing new in kind is needed; what is needed is $\kappa$, which is the curvature of the energy against a *non-uniform* displacement of the carriers, and $E_{\rm PN}$, which must be inferred from $\kappa$ and $K$ rather than read off a slide. Neither is measured here.

What is outside is not a regime but a kind of quantity. A resistive conductivity $\sigma = ne^2\tau/m$ needs a momentum relaxation time from electron–phonon scattering, and that needs a current as an object, electron dynamics, and irreversibility; a minimisation over real densities has none of the three, at any barrier height. And band conduction specifically is absent in kind rather than by degree: non-overlap admits no extended states, so there is nothing for a band to be made of, and no choice of stiffness produces one. Neither of these tells us what the unpinned state of §5 is. They say only that if it conducts, it does not do so by filling bands, and that this framework will not supply the rate at which it does.

## 8 Status

The corrugations above are rigid-slide values, and as §6 sets out they are upper bounds: the kink is the cheap path and its barrier is not measured here. The states entering each difference carry Euler–Lagrange residuals of about $4 \times 10^{-3}$, so they are not fully relaxed; the stiffened runs halve the timestep and are less relaxed than the others; and the response to stiffening rests on two values of the coefficient rather than a curve.

**The standoff is imposed.** In Model II the radius $c_r$ separating cores from carriers is prescribed, not found. It is a constraint on the minimisation, and an expensive one: at fixed particle content the energy falls steeply as $c_r$ is reduced, by of order ten electron-volts per electron over the range examined, so the configuration is held open against a strong preference to collapse. Released — the same system with that interface free — the carrier shell falls onto the axis, and every starting standoff returns the same energy. So the cable is a stated arrangement whose conduction properties are measured, not a configuration RealQM produces of itself. Whether the framework has a self-consistent standoff, and what it costs, is the open question this article does not answer.

**The chain's exclusion rule.** Model I uses a pseudopotential of radius $r_c$, and the rule for excluding a domain from a core is not settled: excluding a domain from its own core alone under-excludes, excluding it from every core over-confines, and the two bracket the answer over more than an order of magnitude rather than fixing it. The repair is not a better rule but a system that needs none — a chain of hydrogen atoms, kernel charge $+1$ with one electron per site and $r_c = 0$, where the carrier is kept off its neighbours by non-overlap alone, exactly as in the cable. That calculation is not reported here.

**Resolution.** Every number above is for the stated system at the stated mesh, with the nuclear cutoff held at an absolute radius above $2h$ so that it is a property of the model rather than of the grid. Whether the same numbers survive a different discretisation, or a different implementation of the same equations, is not established here and is left to further work.

What does not depend on any of this is the instrument. Equation (3) cancels ends and junctions by construction and tests itself against a third window, and the controls throughout are of the kind that can fail: translation by one spacing must return the same energy to the last digit, and domains related by a symmetry must have equal eigenvalues.

## 9 Conclusion

For a chain of atoms, the valence electron occupying its own core, RealQM gives a corrugation of order an electron-volt for the electron pattern to advance one site, and a polarisability extensive in the number of electrons. The charge is fixed per site and stays there: an insulator.

For a cable of the same ions with the carriers standing off their cores, it gives about 240 meV — the range over which the transport of real semiconductors is thermally activated. The interaction is the same, the energy is the same, the ion is the same. Only the arrangement of the charge differs, and the barrier differs with it by a factor of four.

For the same cable again, with the carriers stiffened — the coefficient of the kinetic energy doubled — it gives about 24 meV. That is thermal energy at room temperature, and a barrier at $kT$ does not confine: the pinning is gone and the charge is free to move. What state the system is then in the measurement does not say, only that it is no longer held.

That progression is the answer, and what makes it structural rather than incidental is that the barrier is set less by the ion lattice than by how strongly the carrier distribution resists being modulated by it. RealQM's free interface resists very little, which is why the pinning is there in the first place and why a little stiffness is enough to remove it.

Taking the corrugation as the activation energy and closing the prefactor with an attempt frequency built from the same landscape and an ionic mass puts the two arrangements at $10^{-12}$ and $10^{0}\ \mathrm{S\,m^{-1}}$, glass and germanium, with the attempt frequencies coming out at lattice values unasked. Those numbers carry three imports — the mass, since RealQM has none; the carrier density; and the curvature's assumed shape — and they are lower bounds, since charge moves by kinks rather than by rigid slides. A resistive conductivity is outside altogether: its relaxation time needs a current as an object and irreversible dynamics, and a minimisation over real densities has neither.

# A Computational details

All calculations use the RealQM solver `molecule.js` driven by `chain_slide.html` (WebGPU, single GPU). Equation (1) is minimised by imaginary-time relaxation of the densities on a uniform Cartesian grid, with the partition specified by a cell-ownership map.

**The code, and running it.** Everything is at

`https://github.com/Claes542/RealMolecule`

and runs *in a web browser* with nothing to install, compile or configure: the solver is JavaScript and WGSL, the arithmetic is done by WebGPU on whatever GPU the machine has, and a run is started by opening a page. A WebGPU-capable browser is required — Chrome or Edge at present. Every configuration in this article is a query string, so each measurement below is reproduced by following one link. The reference state and the two windows that give the 240 meV of §4:

```
https://claes542.github.io/RealMolecule/chain_slide.html?n=16&a=4.5&rc=0&rsing=0.5&cps=8&
wrap=1&coax=1&ncore=2&chom=1&cr=2.4&d=0
```

and the same with `&swin=4&d=0.5`, `&swin=8&d=0.5`, `&swin=12&d=0.5`; the stiffened runs of §5 add `&kins=1`. Each run relaxes until its energy stops changing and then writes its total energy, its Euler–Lagrange residual and its per-domain eigenvalues into `localStorage` under `realqm_resid`, keyed by the full configuration, so the numbers in the tables below are read out rather than transcribed from the screen. The parameters are named in the source at the top of `chain_slide.html`. For every corrugation the partition is *frozen*: the energy of a stated configuration is evaluated, which is legitimate for configurations related by a symmetry of the lattice and is why translation by one spacing is used as the control. Energies are reported in hartree and converted at 1 Ha = 27211 meV.

**Model I: chain of atoms.** $N = 7$ model atoms in a row, nuclear charge $+1$, one valence electron each, wrapped labelling so that translation by one spacing is an exact relabelling. Two geometries were used: $r_c = 1.0\,a_0$ at spacings $a = 3.5$ to $8.0\,a_0$ for the polarisability and the density trend, and $r_c = 1.93\,a_0$ at $a = 5.5\,a_0$, $h = 0.448\,a_0$ for the corrugation, $r_c$ in the second being fixed by the isolated atom's ionisation energy rather than chosen. Polarisability from the induced dipole at two field strengths with the partition frozen, at $N = 3, 5, 7$: $\alpha = 27.4N\,a_0^3$, flat to 1% per atom. Corrugation from $E(0)$ and $E(\frac{1}{2})$; on the $a = 5.5$ chain, $E(0) = -1.631277$ Ha throughout, with $E(\frac{1}{2}) = -1.839$ Ha under own-core exclusion (808 meV per electron) and $-1.617$ Ha under exclusion from every core (54 meV). These bracket the answer; the electron-volt scale quoted in §3 is the own-core end, and the ∼0.8–1.5 eV range across spacings from 3.5 to $8.0\,a_0$ is measured on the $r_c = 1.0$ chain.

**Model II: cable of ions surrounded by electrons.** $N = 16$ sites at $a = 4.5\,a_0$ on the axis, kernel charge $+3$, one core domain per site carrying two electrons and confined within $c_r = 2.4\,a_0$ of the axis, one valence electron per site outside that radius: 32 domains, neutral, $r_c = 0$, so there is no pseudopotential and the carrier is excluded from the core region by non-overlap alone. Mesh 8 cells per spacing, box $= (N-1)a + 10 = 77.5\,a_0$, grid $138^3$, $h = 77.5/138 = 0.5616\,a_0$; the box is the chain span plus a fixed pad and so does not depend on the mesh.

**The nuclear cutoff, and a correction.** A bare $-Z/r$ well cannot be represented on a grid, so it is clamped at a radius $R_{\text{sing}}$. The solver sets $R_{\text{sing}} = \max(2h,\, r_{\text{sing}})$ with $r_{\text{sing}}$ a requested absolute value, and the requested $0.5\,a_0$ used here is smaller than $2h$ at every mesh employed — 1.12, 0.90 and $0.75\,a_0$ at 8, 10 and 12 cells per spacing. The cutoff was therefore $2h$ throughout, tied to the grid rather than absolute, and refining the mesh sharpens the ion instead of merely resolving it. An earlier version of this appendix stated that the clamp fixed the potential as the same object at every mesh; that is not what the code does, and any comparison across meshes made under that assumption compares different ions rather than one ion at different resolutions. To hold the potential fixed, $r_{\text{sing}}$ must be set above $2h$ at the finest mesh in use. Only the shell slides. Energies at $d = \frac{1}{2}$ with $M$ middle carriers displaced, against the undisplaced reference $E = -160.133661$ Ha:

| $M$ | 4 | 8 | 12 |
|---|---|---|---|
| $E$ (Ha) | $-160.352807$ | $-160.312817$ | $-160.281810$ |
| $\Delta E$ (meV) | $-5963$ | $-4875$ | $-4031$ |

Equation (3) on $(4, 8)$ gives $\varepsilon_{\text{bulk}} = 272$ meV and $\varepsilon_{\text{junc}} = -3526$ meV per junction; on $(8, 12)$ it gives 211 meV. The linearity test predicts $\Delta E(12) = -3787$ meV against $-4031$ measured, a 6% miss, and the spread between the two estimates is quoted as the uncertainty: $240 \pm 30$ meV. The $(4, 8)$ pair is the cleaner, both its junctions lying well inside the chain, where at $M = 12$ only two sites remain on either side.

**Stiffened carriers.** Identical geometry, mesh and ion lattice, with the coefficient of $|\nabla\psi_i|^2$ raised from $\frac{1}{2}$ to 1 for the shell domains only, the cores keeping $\frac{1}{2}$. The coefficient enters three places — the imaginary-time step, the $H\psi$ diagnostic and the energy sum — and all three must carry it or the reported energy does not belong to the state produced. The explicit step requires the diffusion number below 1/6, so the timestep is halved in proportion; the stiffened runs are correspondingly less relaxed. $\Delta E(4) = -8447$ meV, $\Delta E(8) = -8352$ meV, giving $\varepsilon_{\text{bulk}} = 24$ meV per carrier.

**The rate estimate of §6.** Equation (4) evaluated at $T = 300$ K ($kT = 25.9$ meV) with $a = 4.5\,a_0 = 2.381$ Å, $n = 1/a^3 = 7.4 \times 10^{28}$ m$^{-3}$, $M = 28\,u$, and $K = 2\pi^2 V_0/a^2$ from the assumed sinusoid, giving $\nu = (1/a)\sqrt{V_0/2M}$:

| $V_0$ (meV) | $K$ (eV Å$^{-2}$) | $\nu$ (Hz) | $\sigma_0$ (S/m) | $e^{-V_0/kT}$ | $\sigma$ (S/m) |
|---|---|---|---|---|---|
| 1000 | 3.48 | $5.5 \times 10^{12}$ | $1.4 \times 10^5$ | $1.6 \times 10^{-17}$ | $2.3 \times 10^{-12}$ |
| 240 | 0.84 | $2.7 \times 10^{12}$ | $7.0 \times 10^4$ | $9.3 \times 10^{-5}$ | 6.5 |
| 24 | 0.084 | $8.5 \times 10^{11}$ | $2.2 \times 10^4$ | 0.395 | $8.8 \times 10^3$ |

Taking $M = m_e$ instead of an ionic mass raises $\nu$ to $6.1 \times 10^{14}$ Hz and $\sigma$ to $1.5 \times 10^3$ S m$^{-1}$ on the middle row; that frequency is two orders above any lattice vibration, which is the argument for the ionic choice quite apart from the physical one. The $\pm 30$ meV uncertainty on the middle row is a factor 3.2 in $\sigma$.

**Controls applied.** Translation by one full period must return the same energy: satisfied to nine digits ($E(0) = E(N)$ exactly) on the wrapped geometries. Repeat runs of one configuration agree to nine digits, so the solver is deterministic and any discrepancy is attributable to the model or the input. Domains related by a symmetry must have equal eigenvalues: this is what exposed the terminal domains of a finite cable binding an electron-volt harder than the interior, and hence the necessity of (3). Euler–Lagrange residuals are $4 \times 10^{-3}$ for the chain and 2–$3 \times 10^{-2}$ for the cable.

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